03 21826 Autumn 2025
LH The Life and Death of Stars

Lecture 15

(JCD 11)

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Previous lecture

- Star formation happens from gravitational contraction of molecuar gas clouds
- The Jeans mass is defined as the mass a cloud should at least have for collapse to happen

$$M_J = 2 \times 10^{-11}\,\mathrm{M}_\odot \left(\frac{T}{\mathrm{K}}\right)^{3/2} \left(\frac{\rho}{\mathrm{g\,cm^{-3}}}\right)^{-1/2}$$

- Stars can only start fusing hydrogen if they are more massive than $0.08\,\mathrm{M}_\odot$
- Stars initially evolve via a Hayashi track and then the Henyey line

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Outline

Zero Age Main Sequence

Core H burning

Main sequence lifetime

Summary

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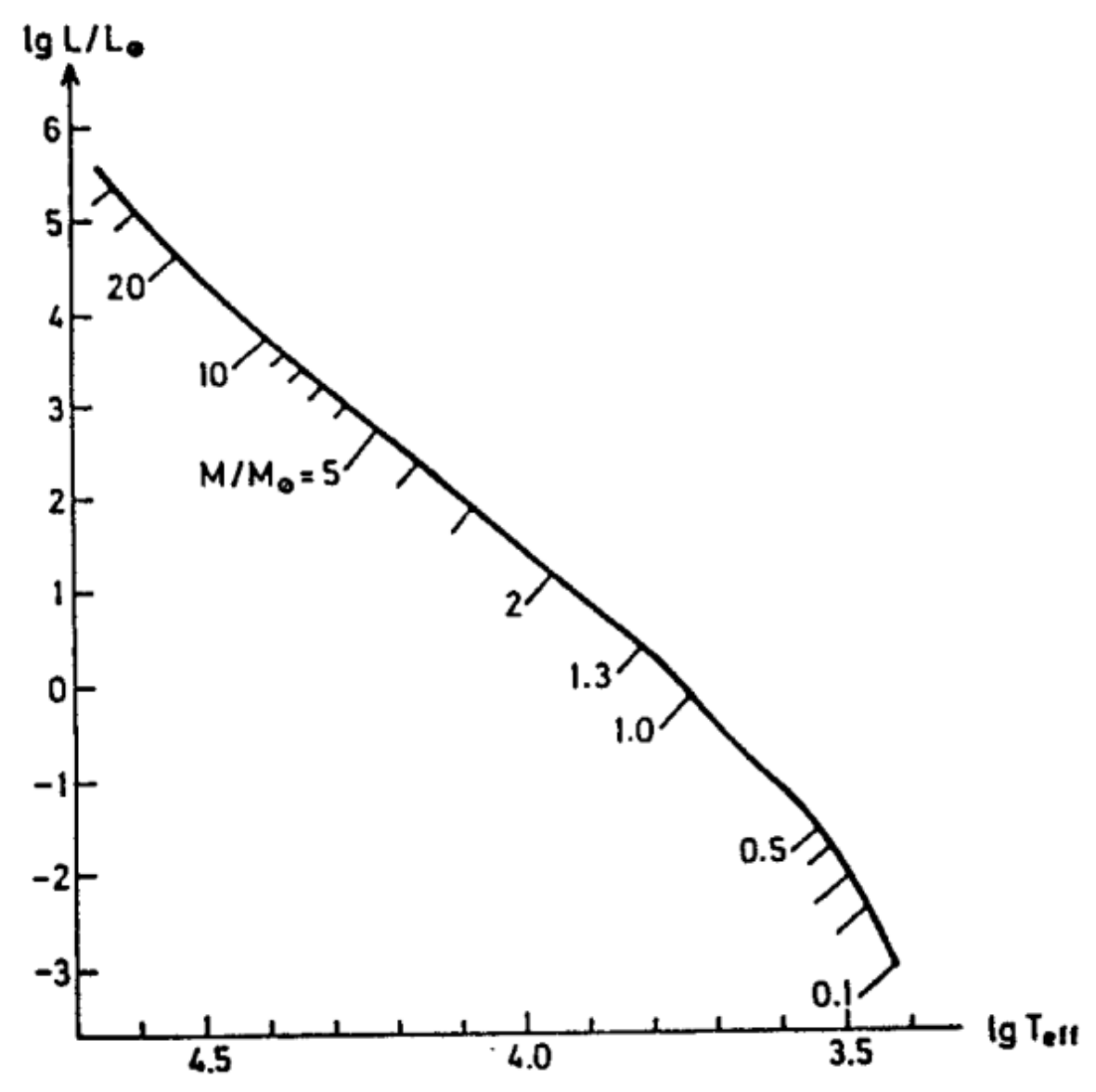
Zero Age Main Sequence = ZAMS

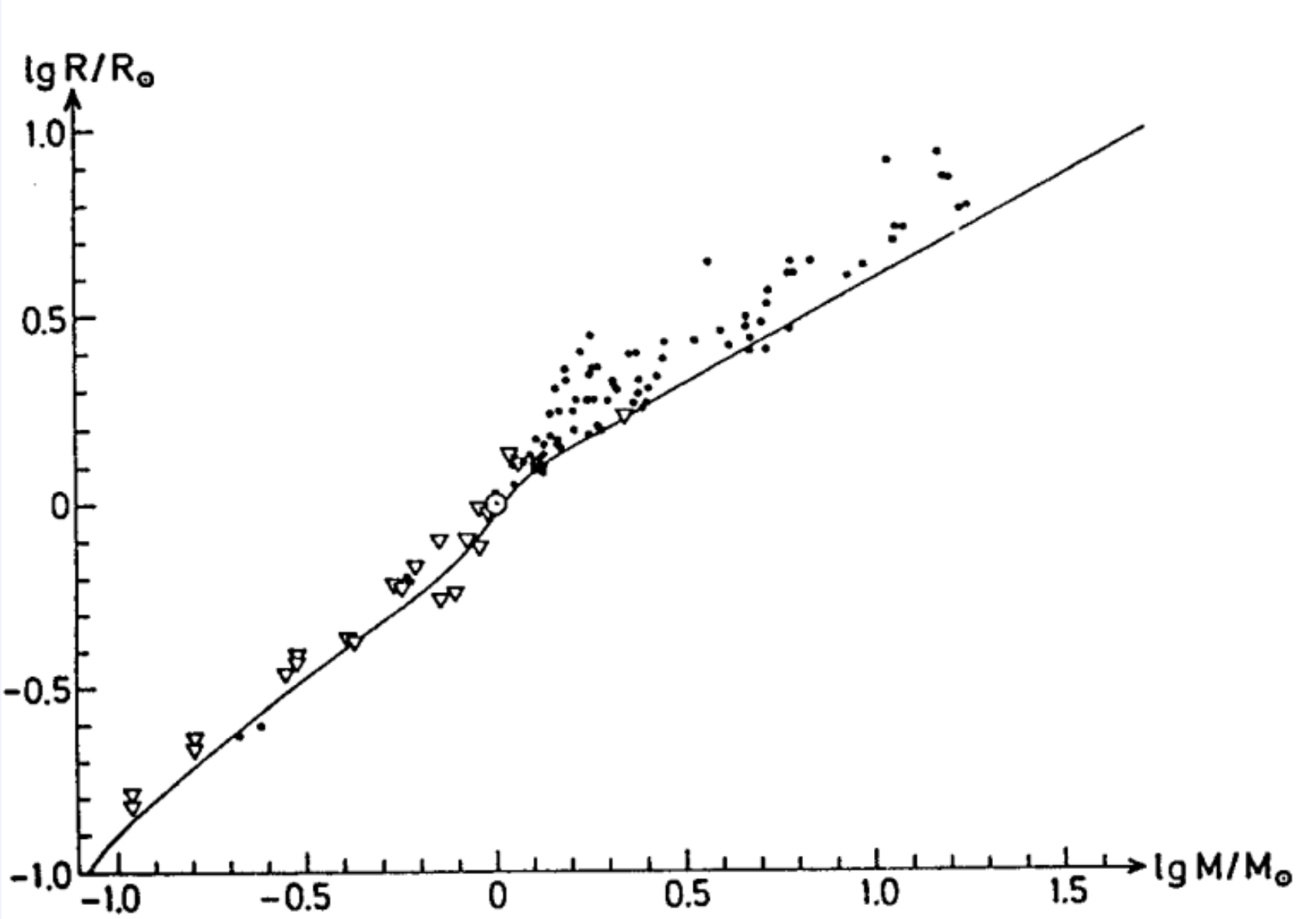
- ZAMS is loosely defined as the region where a star has settled once it starts fusing hydrogen in its core
- The location thus depends mainly on initial mass and initial chemical composition (or mean molecular weight)
- The stellar properties of mass, radius, luminosity and effective temperature are all closely related on the ZAMS
- In upcoming figures, young binary star properties are compared to results from numerical relations

Luminosity, effective temperature, and mass



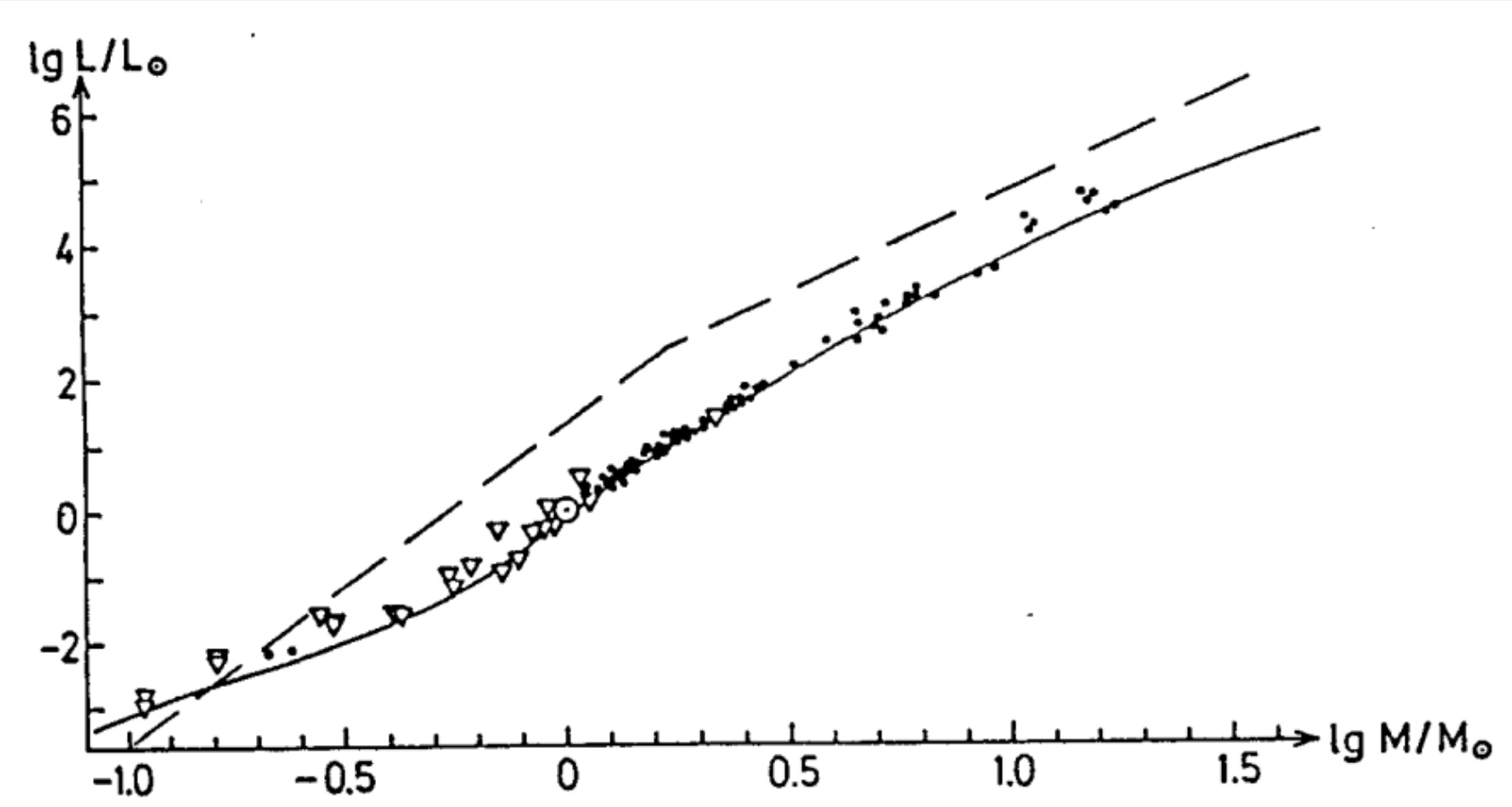
Credit: JCD, Fig 11.1

Mass and radius



Credit: JCD, Fig 11.2

Mass and luminosity



Credit: JCD, Fig 11.3

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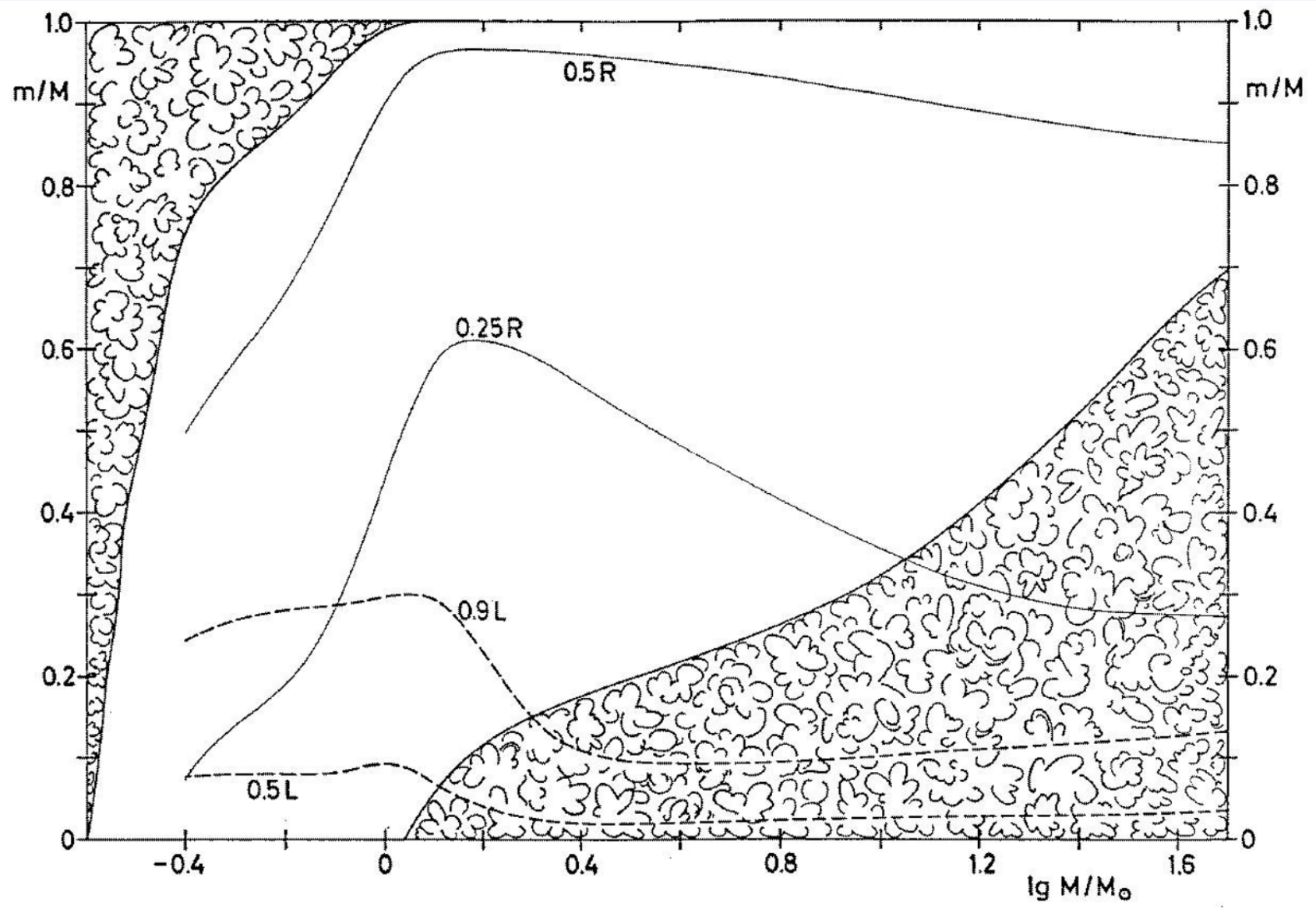
Different regions in ZAMS stars



Credit: JCD, Fig 11.4

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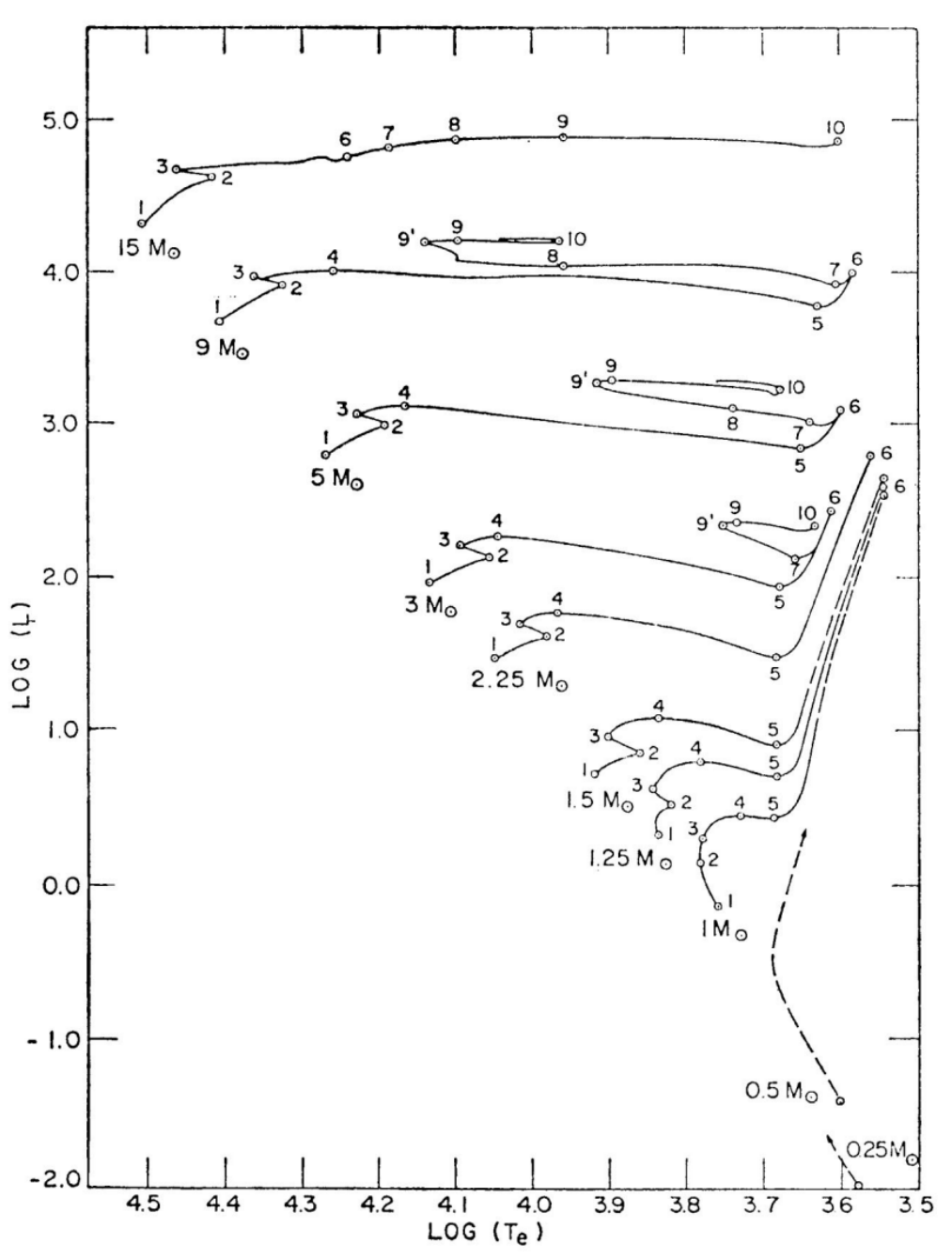
Evolution during core H burning

- Once a star has reached the ZAMS and starts burning hydrogen in the core, its location in the HR diagram will change slowly
- When hydrogen stops burning in the core, a star has reached the "TAMS" = Terminal Age MS
- Generally, all stars will increase in luminosity during core H burning
- Verify this statement by thinking of what happens in a star during core H burning: X will decrease, hence μ increases, hence $\rho\, T$ increases (as P can not decrease), hence ϵ increases, hence L increases
- This is consistent as well with the mass-luminosity relations we found
- What happens with radius and effective temperature (and why) is less well understood. Generally radius will increase. Effective temperature will either increase (low mass) or decrease (high mass)



ZAMS = 1, TAMS = 2 (Credit: JCD, fig 11.5)

Remember the Gaia colour-magnitude diagram

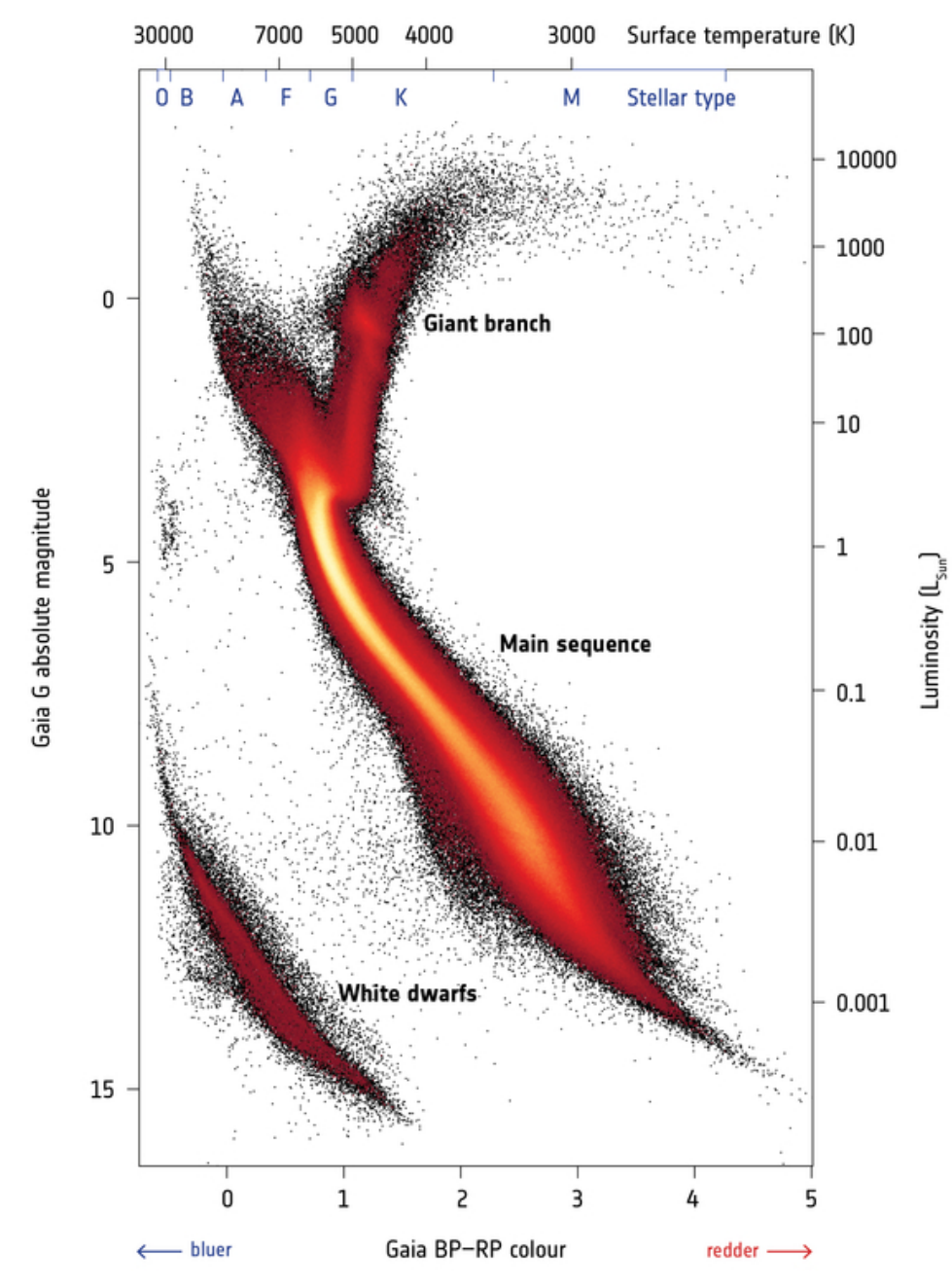
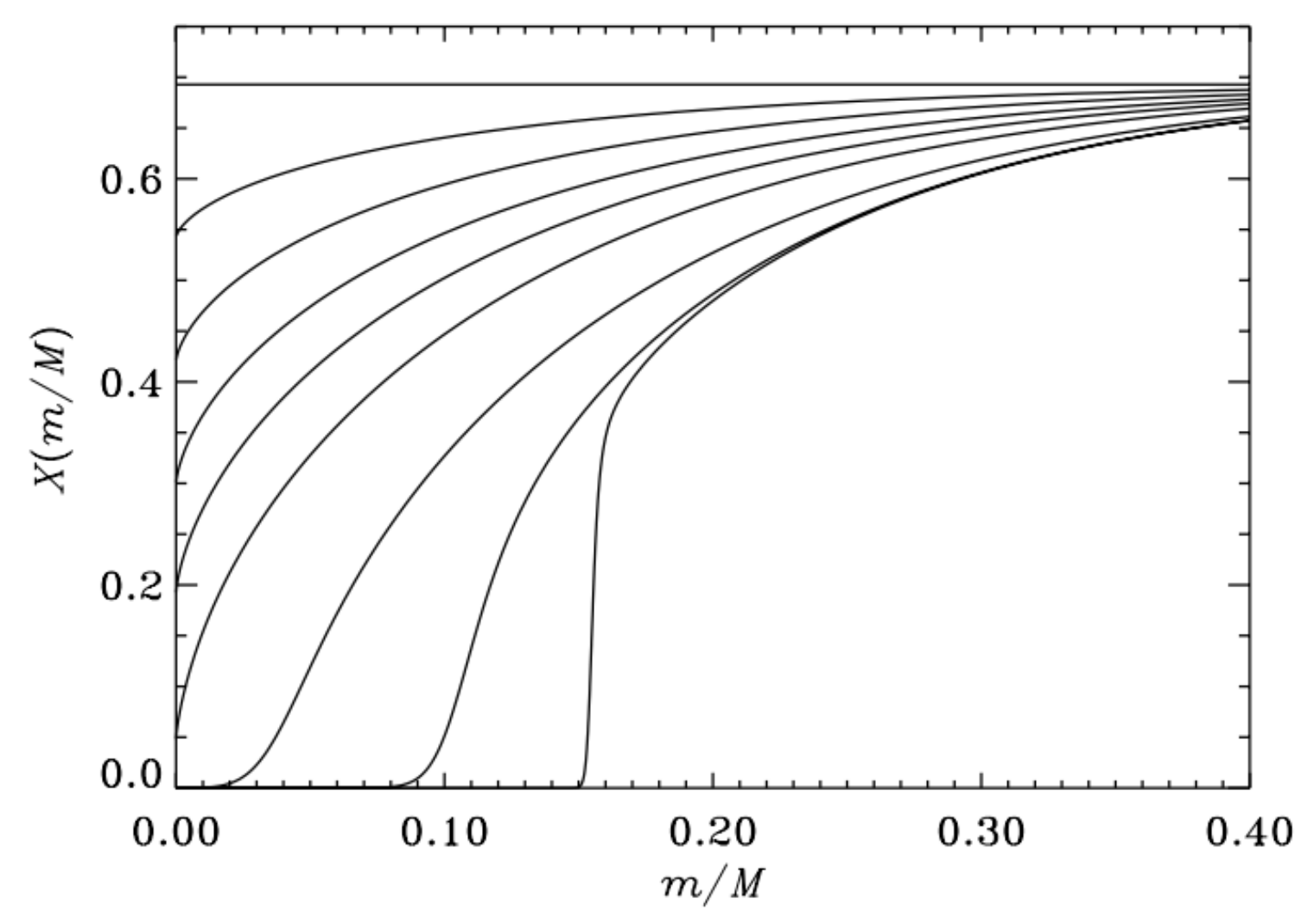


Image credit: ESA

Hydrogen abundance changes

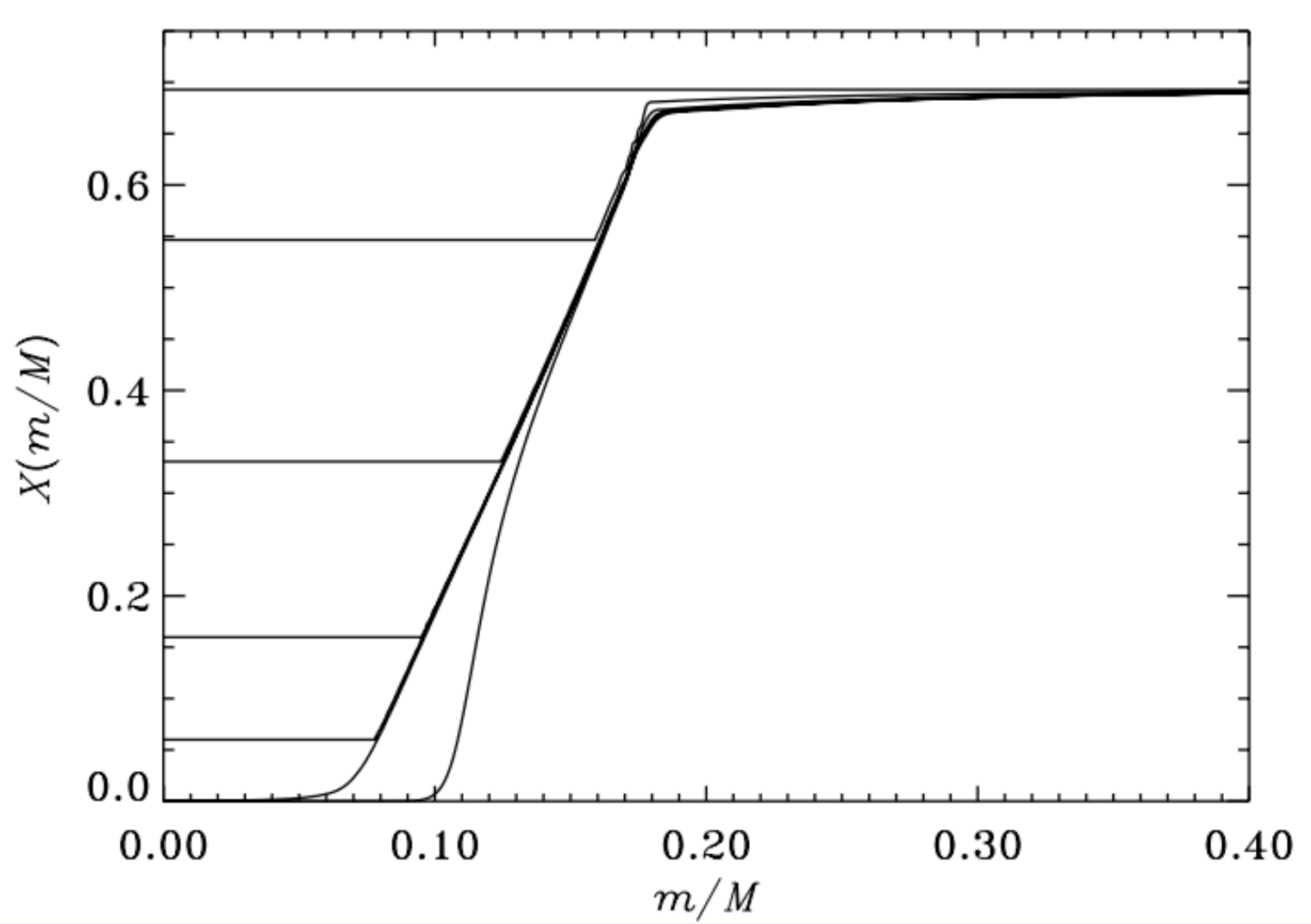
- What happens to the hydrogen abundance (or X) during core H burning depends whether the burning takes place in a convective or radiative zone
- Radiative zone: X goes down in the core and increases gradually further out. Full exhaustion of hydrogen happens first precisely at the centre
- Convective zone: Optimal mixing of elements, hence uniform decrease of X in the zone

Change of X in radiative core of 1 solar mass star



Credit: JCD, fig 11.6 - check caption in full notes!

Change of X in convective core of 2.5 solar mass star



Credit: JCD, fig 11.7 - check caption in full notes!

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Main sequence timescales

- MS lifetime is determined by the amount of hydrogen available for burning and the luminosity of the star [remember lecture 1]
- We also need the average amount of effective energy release $\bar{Q}_{\rm eff}$
- Defining q_c as the fraction of stellar mass that takes part in hydrogen burning (roughly 10% for the Sun), we find that

$$t_{\rm MS} = \frac{q_c\, X_0\, \bar{Q}_{\rm eff}\, M_*}{4\, A_H\, m_u\, L_*}$$

- Since q_c and $\bar{Q}_{\rm eff}$ are roughly independent of mass and we know mass and luminosity are related by $L_* \propto M_*^{\nu}$ with $\nu = 3 - 5$, we find that

$$t_{\rm MS} \propto M_*^{1-\nu}$$

Main sequence timescales dependence on mass

Mass ($M_\odot$) \ Interval ($i-j$)	(1–2)
15	1.010 (7)
9	2.144 (7)
5	6.547 (7)
3	2.212 (8)
2.25	4.802 (8)
1.5	1.553 (9)
1.25	2.803 (9)
1.0	7 (9)

Credit: JCD

Interval 1-2 relates to Main Sequence - see JCD fig 11.5

$$1.010 \quad (7) = 1.010 \cdot 10^7$$

Summary

- When a star starts burning H, it sits on the ZAMS and evolves in the HR diagram till the TAMS
- During core H burning, luminosity and radius will generally increase. Effective temperature can increase or decrease depending on stellar mass
- The change in hydrogen abundance depends on how radiation is transported: gradual change in radiative zones and uniform change in convective zones
- Main sequence lifetime is decreasing function of mass

NEXT LECTURE: Evolution after the main sequence for moderate mass stars